



1583

Plant studies

Italian botanist Andrea Cesalpino developed a method of grouping flowering plants by their fruits, seeds, and roots in his book *De Plantis Libris XVI* (The Book of Plants XVI).

1546–1601

TYCHO BRAHE

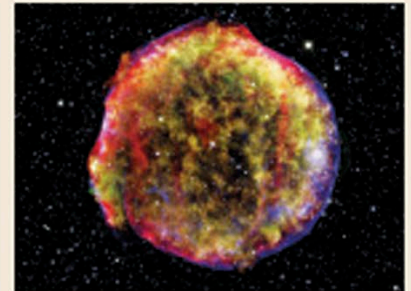
Danish nobleman Tycho Brahe was one of the leading astronomers of his day. Under the patronage of King Frederick II of Denmark, he built a huge observatory on the island of Ven (in Sweden), where he studied the stars and planets. He made all his observations with the naked eye, before the invention of the telescope. As a student, he lost part of his nose in a duel and wore a false metal one for the rest of his life.

Brahe's new star

In 1572, Brahe observed a bright new star in the constellation of Cassiopeia (at the top, labeled "I" on his map). Modern telescopes have revealed that this was an exploding star, or supernova.



Brahe's map of Cassiopeia showing his new star



Modern-day image of SN 1572, believed to be Brahe's supernova

1585

1590

1582

New calendar

Pope Gregory XIII introduced a new calendar to modify the Julian calendar, which had been used in Europe since Roman times. Known as the Gregorian calendar, this calendar calculated the date of the holy festival of Easter more accurately. At first recognized only by Catholic countries, the Gregorian calendar is now used by many countries around the world.



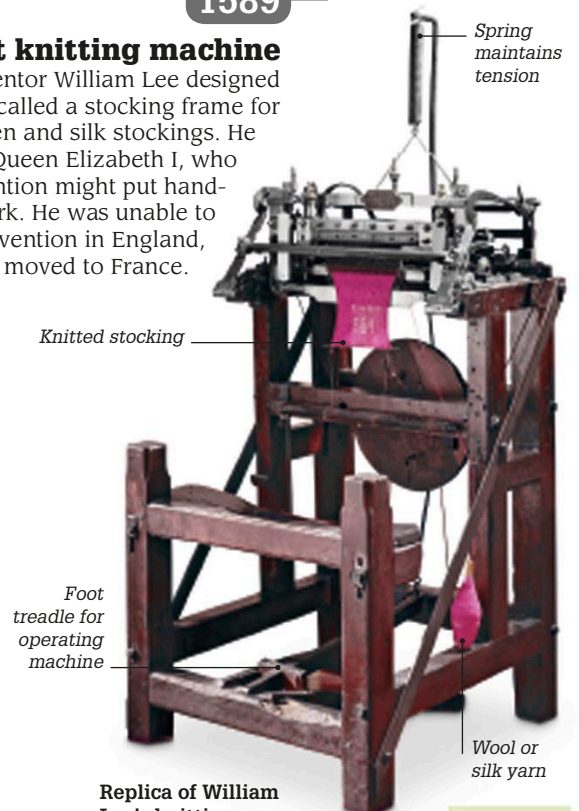
Pope Gregory XIII presides over the discussion to reform the calendar in Rome

Great Britain did not switch to the Gregorian calendar until 1752.

1589

First knitting machine

English inventor William Lee designed a machine called a stocking frame for knitting woollen and silk stockings. He demonstrated it to Queen Elizabeth I, who feared his invention might put hand-knitters out of work. He was unable to patent his invention in England, so he moved to France.



Replica of William Lee's knitting machine